JKF CMMS

THIAGARAJAR COLLEGE OF ENGINEERING

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Industry Maintenance System

ABSTRACT

This project presents a web-based Computerized Maintenance Management System for monitoring industrial machines, maintenance work orders, spare resources and safety records. The system replaces scattered manual tracking with a centralized dashboard where Admin, Manager and Technician users can view machine status, record breakdowns, assign work and analyze maintenance performance.

A predictive maintenance module extends the CMMS by using historical failure and repair logs to estimate MTBF, MTTR, failure probability and risk level. The solution combines Node.js, Express, EJS, SQLite and an XGBoost-based Python microservice to support faster maintenance decisions and improved machine availability.

BACKGROUND

Manufacturing equipment such as mixers, mills, extruders, presses, conveyors and cooling systems require continuous monitoring to avoid unexpected downtime. Traditional maintenance practices depend heavily on delayed reporting and manual stock checks, which can increase repair time and reduce visibility.

A CMMS improves reliability by connecting machine health, technician activity, spare material usage and safety incidents in one application. Predictive analytics further helps identify machines that may need attention before a major breakdown occurs.

OBJECTIVES

- Digitize machine and maintenance records.
- Track work orders and technician tasks.
- Monitor MTTR, MTBF and machine status.
- Predict maintenance risk using ML models.

METHODS

System Design: The application follows a web-based architecture with an Express server, EJS pages, Bootstrap styling and SQLite persistence.

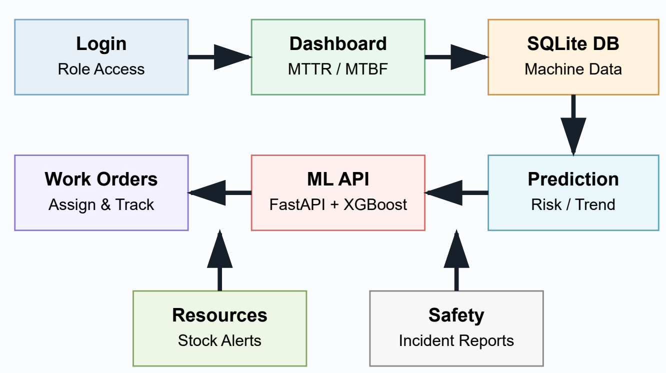
Authentication: Users sign in through role-based accounts for Admin, Manager and Technician permissions.

Maintenance Records: Work orders store machine, title, priority, assigned technician, failure time, start time, end time and status.

Resource Control: Materials include current quantity, threshold value and machine-wise usage to support spare planning.

Predictive Engine: The Node.js app calls a Python FastAPI service. XGBoost models process rolling-window features from the latest ten logs.

ARCHITECTURE ILLUSTRATION



Data flow from role login to dashboard, database, work orders, inventory and predictive risk output.

RESULTS

10,000 Generated maintenance records

50 Simulated machine IDs

3 ML model outputs

Dashboard Monitoring: The dashboard displays MTTR, MTBF, total machines, active machines and breakdown machines for quick decision-making.

Work Order Control: Managers and admins can create maintenance work orders, assign technicians and update progress from pending to in-progress or completed.

Machine Intelligence: Each machine detail page can request predicted MTBF, predicted MTTR, failure probability, risk level and trend from the ML service.

Inventory Visibility: Low-stock spare materials are highlighted automatically, and resource reports can be exported for planning.

Operational Safety: Safety reports are stored with employee name, description and date, helping the team maintain incident history.

MODEL FEATURES

Feature	Purpose
Average MTBF	Measures recent uptime behavior before failure.
Average MTTR	Measures recent repair duration pattern.
MTBF Trend	Detects whether machine reliability is improving or decreasing.
MTTR Trend	Identifies changes in repair complexity.
MTTR Variance	Captures inconsistency in repair time.

OUTPUT CLASSES

LOW RISK | MEDIUM RISK | TREND

CONCLUSIONS

The proposed Smart CMMS improves maintenance visibility by integrating machine details, breakdown history, technician assignments, resource stock and safety reporting into one web application. It supports faster response during machine failure and helps managers review maintenance performance using MTTR and MTBF values.

The predictive module adds forward-looking support by estimating failure probability and risk level from recent logs. This enables preventive action, better spare planning and reduced production downtime. The modular structure also allows future extension with live sensor readings, notification alerts and richer reporting.

ADVANTAGES

- Centralized maintenance history for all machines.
- Role-based access for secure task handling.
- Real-time dashboard view of operational condition.
- Automated low-stock indication for spare materials.
- ML-supported prediction for proactive maintenance planning.

FUTURE SCOPE

- Connect IoT sensors for live vibration, temperature and load data.
- Add email/SMS alerts for high-risk machines and low stock.
- Include charts for monthly breakdown trends and technician workload.
- Deploy the ML service and web app together for production use.